Case 19: Transient WUI heat flux

1 Verification purpose

This coupled-process case verifies the transient wildland–urban interface (WUI) heat-flux pathway that combines the building-spread ellipse, a piecewise design-fire heat-release curve, direct-flame-contact (DFC) transfer, radiative transfer, and transient raster output. It exercises the `WUI` namelist, WUI state updates in ELMFIRE level-set formulation, and transient output branches in ELMFIRE output procedure. It can expose wind-angle, ellipse-geometry, time-curve, cell-normalization, heat-partition, or output-indexing errors. Wildland fuel, crown fire, spotting, and suppression are not decision observables. This is algorithm verification against a separately evaluated documented formulation, not physical validation. The question is whether the prepared case retains exactly one active WUI source, all required transient heat-release rate (HRR) per unit area (HRRPUA), DFC, and radiation stacks exist, and each energy-normalized error is no greater than 0.005.

2 Mathematical formulation and numerical implementation

Wind speed is converted from mph to SI as $V = 0.447 V_{\text{mph}}$. For configured building area A, separation D, and wind proportionality W_p, the selected Hamada-style regression produces downwind, upwind, and sidewind distances D_d, D_u, D_s. Coefficients differ for $V < 10$, $10 \le V \le 17.3$, and $V > 17.3$ m s^{-1}; this case uses 15 mph $= 6.705$ m s^{-1}, so it exercises the low-wind linear branch. The translated ellipse has

$$a = (D_d + D_u)/2, \qquad e = \min(a/2, a - D_u), \qquad b = D_s/\sqrt{1 - (e/a)^2}.$$

For target direction θ_f, its boundary distance is

$$R_e(\theta_f) = \frac{0.3 D_d (a - e)}{b^2} \frac{b^2}{a - e\cos\theta_f}, \quad \theta_f = \text{atan2}(y, x) - \pi(270 - WD)/180,$$

which preserves the meteorological wind-from convention.

For burning time τ, the design fire rises linearly to H_p at t_e, remains at H_p through t_d, decays linearly to zero at t_c, and is zero thereafter:

$$H(\tau) = \begin{cases} H_p \tau / t_e, & 0 \le \tau \le t_e, \\ H_p, & t_e < \tau \le t_d, \\ H_p(\tau - t_c)/(t_d - t_c), & t_d < \tau \le t_c, \\ 0, & \text{otherwise.} \end{cases}$$

Cell coverage is $C_D = \text{clip}[(R_e + \Delta/2 - R)/\Delta, 0, 1]$. With configured ellipse scale $\eta = 0.5$, $C_H = \Delta^2/[\pi(\eta a)(\eta b)]$, burnable fraction C_b, absorptivity α, radiation-annulus fraction F_r, and effective separation R_{eff},

$$q''_{\text{DFC}} = C_b C_D H C_H, \qquad q''_{\text{rad}} = \frac{0.3 C_b \alpha F_r C_H H \Delta^2}{4\pi R_{\text{eff}}^2}.$$

The case-specific reference functions evaluate these equations independently at the timestamps emitted by ELMFIRE.

3 Derivation of expected behavior

The configured constants are $A = D = 10$ m, $W_p = 1$, $\eta = 0.5$, $\Delta = 20$ m, $WD = 0°$, $H_p = 400\,\mathrm{kW\,m^{-2}}$, $t_e = 300$ s, $t_d = 3900$ s, and $t_c = 4200$ s. The generated adjustment field is uniformly zero, so the center WUI structure cannot ignite a secondary level-set source. Because the current executable generic stalled-front shortcut does not account for an active WUI design-fire curve, one point ignition is configured at the center for $t = 5001$ s, strictly after the 5000 s stop. It is never applied; its only role is to keep that shortcut inactive while the prescribed curve is sampled. Substitution into the low-wind regression fixes one time-independent ellipse. At each timestamp, the piecewise equation fixes HRRPUA; the ellipse and coverage equations then fix each cell in the 11-by-11 audit window. DFC is confined to covered cells and scales with H; radiation occupies the outer annulus and decays with R_{eff}^{-2}. This reference depends on configured constants and timestamps, not on simulated heat-flux values.

The analytical history is evaluated with the generated constants $A = D = 10$ m, $W_p = 1$, $\eta = 0.5$, $H_p = 400\,\mathrm{kW\,m^{-1}}$, and $(t_e, t_d, t_c) = (300, 3900, 4200)$ s. Thus the prescribed triangular heat-release history is

$$H(t) = \begin{cases} 400t/300, & 0 \le t < 300, \\ 400, & 300 \le t < 3900, \\ 400(4200 - t)/300, & 3900 \le t \le 4200, \\ 0, & t > 4200, \end{cases}$$

so, for example, $H(150) = 200$, $H(300) = H(3900) = 400$, $H(4050) = 200$, and $H(4200) = 0\,\mathrm{kW\,m^{-1}}$. For the generated 15 mph wind, $U = 15(0.44704) = 6.7056\,\mathrm{m\,s^{-1}}$. Substitution in the case's UCB ellipse equations gives $a = 76.2593$ m, $b = 24.8916$ m, $e = 35.6487$ m, downwind reach $D_d = a + e = 111.9080$ m, upwind reach $D_u = a - e = 40.6106$ m, and lateral reach $D_s = 22.0044$ m. Each reported history error uses

$$E_1(q) = \frac{\sum_k |q_{\mathrm{sim},k} - q_{\mathrm{ref},k}|}{\sum_k |q_{\mathrm{ref},k}|},$$

with q equal to heat release rate, dry-fuel consumption, or radiation. The exact-reference substitution is $E_1(q) = 0$, and each component is accepted when $E_1(q) \le 0.005$.

4 Verification metrics and acceptance criteria

For each observable X,

$$E_1(X) = \frac{\sum_{t,i,j} |X_{\mathrm{ELMFIRE}} - X_{\mathrm{ref}}|}{\sum_{t,i,j} |X_{\mathrm{ref}}|}.$$

HRRPUA uses the center cell; DFC and radiation use the documented 11-by-11 window, restored to mathematical $+y$ orientation. Mutually masked, nodata, and nonfinite samples are excluded. Energy-normalized L_1 error is selected instead of pointwise relative error because the reference is exactly zero over much of space–time; division by those zeros amplifies irrelevant roundoff. It detects both amplitude and footprint errors. Required outputs are the time-resolved heat-release rate, convective heat flux, and radiative heat flux. Their sequential output indices are mapped to physical seconds using the unique record of output times. A 0.5 percent limit allows single-precision raster serialization and cell-selection roundoff while remaining much smaller than a one-cell or coefficient error. All stacks and metrics are required, including one terminal record at 5000 s and dumps spanning growth, plateau, decay, and post-decay. Before evaluation, every nonzero HRR raster must contain exactly the center cell; a second active source makes the single-source experiment NOT EVALUABLE. PASS requires that isolation invariant and all $E_1 \le 0.005$. Missing or incompatible outputs are NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Parameter	Value	Units	Isolation role
Domain raster	20×20 cells	–	centered WUI source and audit window
Analysis cell	20	m	matches reference discretization
Wind	15 at 0° from	mph, degree	low-wind ellipse branch
Building area / separation	10 / 10	m	fixes regression geometry
Nonburnable fraction	0	–	removes availability scaling ambiguity
Ellipse HRR scale η	0.5	–	fixes current normalization
Spread adjustment	0	–	prevents all secondary propagation
Ignition	center PHI cell only	–	the sole physical source
Terminal guard	center at 5001	s	after stop; never ignites
Design-fire times	300 / 3900 / 4200	s	rise, plateau, decay
Peak HRRPUA	400	kW m^{-2}	reference amplitude
Initial/maximum timestep	1 / 10	s	whole-second, CFL-limited stepping
Output interval / stop	100 / 5000	s	exact stop-time alignment and all phases

The predicted HRRPUA rises, plateaus, decays, and becomes zero. Pre-run isolation requires exactly one active HRR cell at every nonzero dump; DFC and radiation therefore come only from the center structure. DFC follows the fixed wind-aligned footprint; radiation surrounds it and both fields scale with HRRPUA.

Figure 1 is generated mechanically from the prescribed input fields. It identifies the prescribed physical input field and representative simulation condition; the left panel shows the generated zero spread-adjustment field that enforces source isolation, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

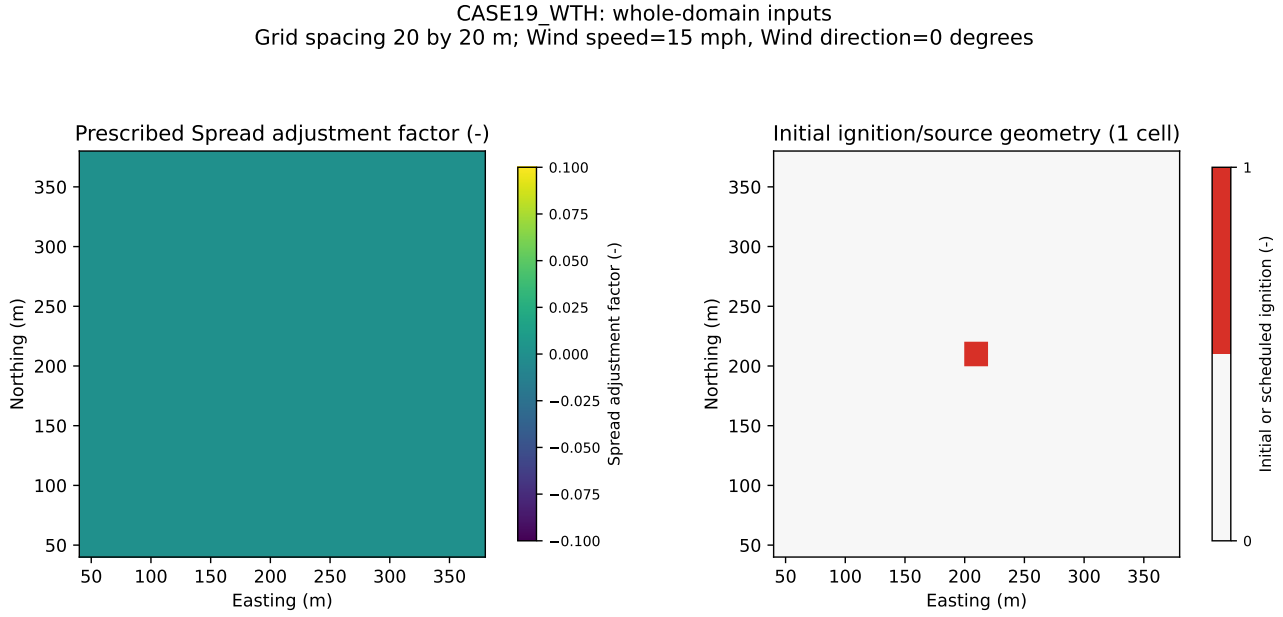


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-specific ELMFIRE GeoTIFFs are used. Current status is **PASS**; 3 of 3 comparison components pass.

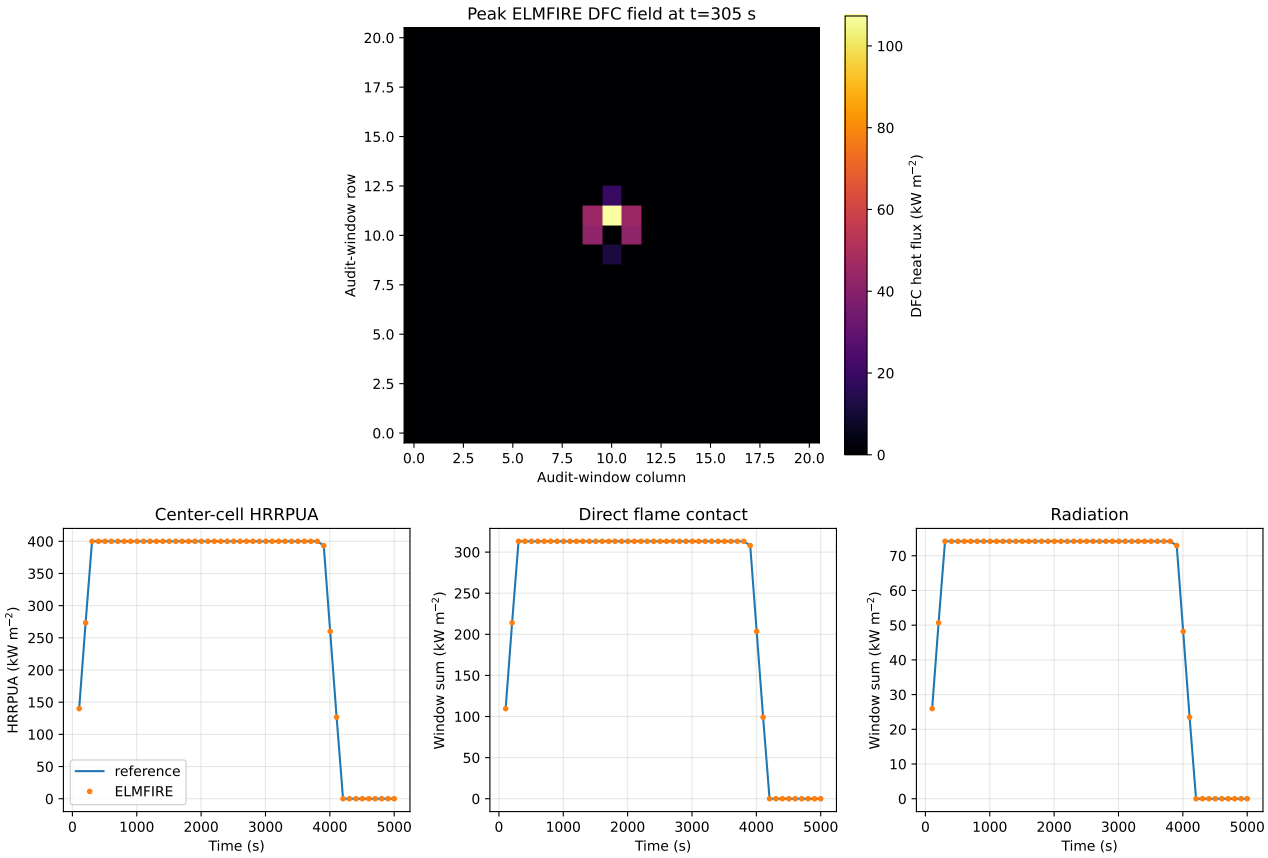


Figure 2: Peak whole-domain DFC heat flux in kW m^{-2} and transient reference-versus-ELMFIRE histories for HRRPUA, DFC, and radiation. All panels are generated from the required case-specific transient GeoTIFF stacks; the histories show every selected output time and the spatial panel shows the maximum-domain-sum DFC output.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
HRRPUA normalized L1 error	0 ($\leq$ 0.005)	1.63091e-09 -	PASS
DFC normalized L1 error	0 ($\leq$ 0.005)	9.02842e-08 -	PASS
radiation normalized L1 error	0 ($\leq$ 0.005)	1.41945e-07 -	PASS

The authoritative decision is **PASS**. The generated table is sourced directly from the case-specific calculated metrics.

8 Technical interpretation

The PASS decision follows the three generated error components after enforcing the single-source invariant. An HRRPUA-only failure implicates burning-time or transient-curve evaluation. A DFC failure with correct HRRPUA implicates ellipse coverage, configured η normalization, orientation, or raster slicing. A radiation-only failure narrows the issue to annulus partitioning, effective distance, or absorptivity. Similar DFC and radiation errors with correct shapes suggest common normalization. A second HRR source is a case-configuration failure and is NOT EVALUABLE; incomplete phase coverage, a nonterminal 5000 s record, or missing output is likewise insufficient evidence, never a pass. The post-stop ignition is a documented workaround for the executable stalled-front shortcut and is outside the verified physical interval.

9 Scope and limitations

This case verifies one deterministic WUI parameter set, one wind-regression branch, one raster resolution, and three transient outputs. It does not verify moderate/high-wind branches, parameter rasters, heterogeneous layouts, structure-to-structure ignition, or empirical validity of the correlations. It supplies implementation-verification evidence only.